Demo Client · DEMO-0005
11 November 2026 · v1.0

Field	Value
Document	Methodology for the independent validation of a credit-risk (PD) model
Template version	model_validation 1.0.0 / proquant-core 0.1.0
Engagement	DEMO-0005 · Demo Client
Model	SME Probability-of-Default Model, SME term loans (synthetic example)
Prepared by	ProQuant Analytics · info@proquantanalytics.com
Run ID(s)	mv_20261111T142054Z_2fe5dd
Status	Final · 11 November 2026

1. Purpose and independence

The validation gives the lender's model-risk function and board an independent opinion on whether a model is fit for its stated use. It is performed by a party that did not develop the model, with the same standards for evidence that apply to the developer. The structure follows common model-risk guidance (conceptual soundness, data, replication and outcome analysis, ongoing monitoring, governance) as applied to probability-of-default models.

2. Inputs and evidence

Data set	Size	Fingerprint (SHA-256)
observations	32,052 rows	`d6f2b749f25ecb63`
model_spec	11 rows	`5c643980e2193c3d`
reported_metrics	7 rows	`1a26e8a36ad177ee`

Each input is fingerprinted with SHA-256 on ingest and recorded in the run audit trail. The production PD is never altered: the model is tested as it runs.

3. Documentation review

The documentation is rated item by item against a checklist grouped in purpose and scope, data, methodology, the developer's performance evidence, implementation and controls, and governance and monitoring. Rating: met = 1, partly met = 0.5, not met = 0, not applicable excluded. Coverage is the score over applicable items. Items marked critical (intended use, definition of default, availability of inputs, assumptions and limitations, reproducibility of the specification, monitoring plan) must be met.

4. Data checks

1. Data-quality checks (schema, missing values, duplicates, outliers) and a log of every cleaning action (exact duplicates dropped; spellings harmonised; impossible values become missing in the validator's working copy; observations with an immature or invalid outcome excluded from outcome analysis).

2. **Timing of inputs.** The lender states when each input is known. An input recorded after the lending decision (post-decision) must not drive a decision model: it describes the future (leakage). Any such input in the model is a finding on its own.
3. **Information value.** IV of every input on the development sample (10 quantile bins, categories as such, 0.5 smoothing); a single input with a very high IV that is not already flagged is a leakage symptom to be explained.
4. Missing rates, number of defaults in development, representativeness (PSI of the PD between development and validation).

5. Replication

1. **Implementation check.** The PD is recomputed from the specification (transformations, caps, documented fills, coefficients) and compared with the production PD (tolerance `replication.pd_tolerance`). The share of loans that differ is reported and diagnosed by the input whose missing values explain the differences.
2. **Re-estimation.** The documented terms are refitted on the development sample by logistic regression; each documented coefficient is tested with z = (estimate - documented) / standard error.
3. **Reported versus recalculated.** Every figure in the documentation (Gini, KS, default rate, number of defaults, observed / predicted ratio) is recomputed on the same sample and compared within tolerance.

6. Outcome analysis

Gini = 2 x AUC - 1, KS and calibration (observed default rate / mean PD, by PD band with binomial p-values and the Hosmer-Lemeshow statistic) on the development, validation (out-of-time) and monitoring samples, with a bootstrap 95% interval for the Gini (200 resamples). Standards are configurable.

7. Challenger models

Challenger	Purpose
Independent logistic benchmark	all inputs known at decision time, simple transformations, built from scratch: is the lender's choice of variables reasonable?
Gradient boosting	an upper bound on what the data support
Remediated specification	the lender's own terms without post-decision inputs, coefficients re-estimated: the "what if the finding is fixed" answer

A challenger that is clearly better than the model points to missing structure; a model that is clearly better than every challenger needs an explanation (see leakage).

8. Stability

PSI of the PD and CSI of every input by quarter against the development sample (below 0.10 stable, 0.10 to 0.25 watch, above 0.25 breach); Gini, default rate and calibration by quarter; bootstrap resampling of the development data (coefficient intervals, sign flips, spread of the validation Gini).

9. Sensitivity

Dropping each term and re-estimating (loss of validation Gini); shocking each raw input by +-x% and recomputing the PD with the specification (change of the mean PD and of the ranking); stress scenarios that move several inputs together.

10. Performance by segment

Default rate, mean PD, calibration and Gini by business age, size, industry and region on the validation sample. A segment is flagged when its Gini is below a share of the overall or its calibration departs from the overall calibration.

11. Findings and opinion

Each test is PASS, WARN or FAIL against thresholds agreed at kickoff. A test that is not passed becomes a finding whose severity depends on the test and the result (configurable): **High** - the model should not be relied on until fixed (due in 30 days); **Medium** - remediate within 90 days with a compensating control meanwhile; **Low** - improvement within 180 days; **Observation** - noted. Analyst findings from interviews and documents are added to the log by hand. The opinion follows the worst severity: any High: not fit for use in its current form; any Medium: fit for use with conditions; otherwise fit for use. The report states what the remediated model would deliver.

12. Assumptions, limitations and deviations

Evidence is supplied by the lender; source systems are not audited. Outcomes are analysed for loans whose window has elapsed. The production PD is tested as delivered, not the surrounding IT controls. The opinion is valid at the report date for the stated use. No deviations from the standard methodology.

13. Reproducibility

Each run folder contains `config_used.yaml`, `audit.json` (analyst, timestamps, code version, input fingerprints, output hashes), `run.log`, every result table as CSV and all documents. Random seeds are fixed (bootstrap, challenger). The replication pack reproduces the core quantitative checks outside ProQuant's code, and is tested against the pipeline at build time.

14. References

- Board of Governors of the Federal Reserve System and OCC (2011). *Supervisory Guidance on Model Risk Management* (SR 11-7).
- European Central Bank (2019). *Guide to internal models* and (2024) *Guide on the supervisory review of internal models*.
- Basel Committee on Banking Supervision (2005). *Studies on the Validation of Internal Rating Systems*, Working Paper 14.
- Siddiqi, N. (2017). *Intelligent Credit Scoring*, 2nd ed. Wiley.
- Hosmer, D., Lemeshow, S., & Sturdivant, R. (2013). *Applied Logistic Regression*, 3rd ed. Wiley.
- Efron, B., & Tibshirani, R. (1993). *An Introduction to the Bootstrap*. Chapman & Hall.